Solution: Problem Challenge 1: Flip and Invert an Image

Problem Statement

Given a square binary matrix representing an image, we want to flip the image horizontally, then invert it.

To flip an image horizontally means that each row of the image is reversed. For example, flipping [0, 1, 1] horizontally results in [1, 1, 0].

To invert an image means that each 0 is replaced by 1, and each 1 is replaced by 0. For example, inverting [1, 1, 0] results in [0, 0, 1].

Example 1:

```
Input: [
  [1,0,1],
  [1,1,1],
  [0,1,1]
]
Output: [
  [0,1,0],
  [0,0,0],
  [0,0,1]
]
```

Explanation: First reverse each row: [[1,0,1],[1,1,1],[1,1,0]]. Then, invert the image: [[0,1,0],[0,0,0],[0,0,1]]

Example 2:

```
Input: [
  [1,1,0,0],
  [1,0,0,1],
  [0,1,1,1],
  [1,0,1,0]
]
Output: [
  [1,1,0,0],
  [0,1,1,0],
  [0,0,0,1],
  [1,0,1,0]
]
```

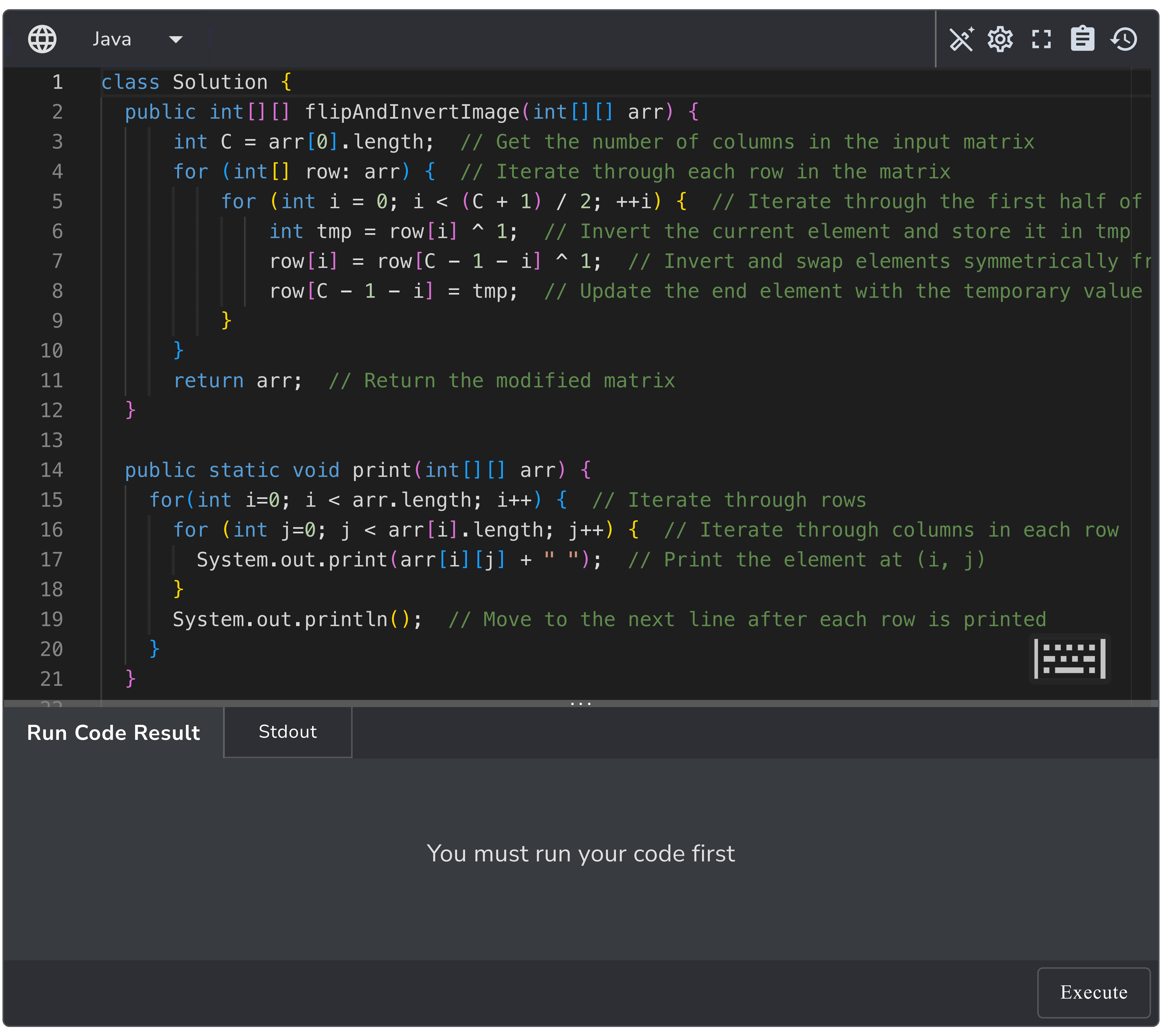
Explanation: First reverse each row: [[0,0,1,1],[1,0,0,1],[1,1,1,0],[0,1,0,1]]. Then invert the image: [[1,1,0,0],[0,1,1,0],[0,0,0,1],[1,0,1,0]]

Solution

1. **Flip:** We can flip the image in place by replacing ith element from left with the ith element from the right.
2. **Invert:** We can take XOR of each element with 1. If it is 1 then it will become 0 and if it is 0 then it will become 1.

Code

Here is what our algorithm will look like:



```java
class Solution {
  public int[][] flipAndInvertImage(int[][] arr) {
      int C = arr[0].length;  // Get the number of columns in the input matrix
      for (int[] row: arr) {  // Iterate through each row in the matrix
          for (int i = 0; i < (C + 1) / 2; ++i) {  // Iterate through the first half of
              int tmp = row[i] ^ 1;  // Invert the current element and store it in tmp
              row[i] = row[C - 1 - i] ^ 1;  // Invert and swap elements symmetrically fr
              row[C - 1 - i] = tmp;  // Update the end element with the temporary value
          }
      }
      return arr;  // Return the modified matrix
  }

  public static void print(int[][] arr) {
    for(int i=0; i < arr.length; i++) {  // Iterate through rows
      for (int j=0; j < arr[i].length; j++) {  // Iterate through columns in each row
        System.out.print(arr[i][j] + " ");  // Print the element at (i, j)
      }
      System.out.println();  // Move to the next line after each row is printed
    }
  }
```

You must run your code first

Time Complexity

The time complexity of this solution is O(n) as we iterate through all elements of the input.

Space Complexity

The space complexity of this solution is O(1).

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